

Alessio Cocchieri

NLP PhD Researcher · UniboNLP · University of Bologna

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About Me

Last-year NLP PhD candidate (ending Nov 2026) at [UniboNLP](#), with 9+ publications at ACL, EMNLP, NAACL, and EACL. I specialize in LLM evaluation and knowledge distillation for low-resource NLP, with applications to high-stakes domains like medicine. Active reviewer for ARR and top-tier ML conferences like NeurIPS.

Education

PhD	University of Bologna , Natural Language Processing - Focus: LLMs, RAG, Information Extraction, Benchmarking, Low-resource NLP - Supervisor: Prof. Gianluca Moro — Research Group: UniboNLP	Bologna, Italy Nov 2023 – present
MSc	University of Bologna , Artificial Intelligence Grade: 110/110 with honors	Bologna, Italy Sept 2021 – Dec 2023
BSc	University of Bologna , Computer Science Grade: 110/110 with honors	Bologna, Italy Sept 2018 – July 2021

Experience

IBM Research , Research Scientist Internship (PhD Intern) Natural Language Processing — LLM Factuality - Created a novel benchmark for multimodal scientific fact-checking, targeting LLM long-form generation - Coordinated a multi-team human annotation pipeline, overseeing annotation guidelines, quality control, and IAA - Conducted systematic analysis exposing critical factuality fragilities in multimodal LLMs across scientific domains	Dublin, Ireland Mar 2026 – May 2026 3 months
IBM Research , Research Scientist Internship (Master Thesis) Natural Language Processing — Named Entity Recognition - Leveraged LLM distillation to improve smaller-sized models for zero-shot NER - Produced two peer-reviewed publications — OpenBioNER (NAACL 2025) and ZeroNER (ACL 2025) - Contributed to the IBM zshot library for zero-shot NER model inference	Dublin, Ireland Mar 2023 – May 2023 3 months

Publications

LLMs (Almost) Never Abstain Under Medical Uncertainty We introduce MedQAbstain, a benchmark for medical abstention under uncertainty, revealing that state-of-the-art LLMs systematically overcommit, rarely abstaining even when the question itself is hidden. <i>Alessio Cocchieri, et al.</i> (ACL 2026 (Main))	Jan 2026
ReMedQA: Are We Done With Medical Multiple-Choice Benchmarks? We show that high LLM accuracy in medical MCQA masks severe inconsistency. We propose novel metrics to evaluate true reliability across MCQA formats. <i>Alessio Cocchieri, et al.</i> (EACL 2026 (Main))	Jan 2026

What do you call a dog that is incontrovertibly true? Dogma: Testing LLM Generalization through Humor

Jan 2025

We introduce Phunny, a novel benchmark using uncontaminated English puns, revealing that LLMs struggle with generalization even on simple tasks, consistently underperforming the human baseline.

Alessio Cocchieri, et al.

(ACL 2025 (Main))

Can Large Language Models Win the International Mathematical Games?

Jan 2025

We introduce MathGames, a novel multimodal benchmark of age-graded math problems from an international competition, showing that frontier LLMs underperform compared to humans, including 11-year-olds.

Alessio Cocchieri, et al.

(EMNLP 2025 (Main))

OpenBioNER: Lightweight Open-Domain Biomedical NER Through Entity Type Description

Jan 2025

We introduce a 110M BERT model that leverages descriptions for zero-shot Biomedical NER, outperforming GPT-4o, specialized LLMs, and GLiNER by up to 10% F1.

Alessio Cocchieri, et al.

(NAACL 2025 (Findings))

To Generate or to Retrieve? On the Effectiveness of Artificial Contexts for Medical Open-Domain Question Answering

Jan 2024

We introduce MedGENIE, the first generate-then-read framework for open-domain medical QA, demonstrating the effectiveness of generated over retrieved contexts and significantly improving LLM RAG performance.

Giacomo Frisoni, *Alessio Cocchieri*, et al.

(ACL 2024 (Main))

Skills

Strong: Python, PyTorch, Hugging Face, vLLM, Scikit-Learn, Git, NumPy, Pandas, NLTK, SpaCy

Proficient: Java, Docker, MySQL, MongoDB, TensorFlow, LangChain, Object-Oriented Programming

Soft Skills: Communication, Problem-Solving, Teamwork, Quick Learning, Leadership, Adaptability, Public Speaking

Languages

Italian: Native speaker

English: Professional fluency

Spanish: Basic proficiency